

Polarization mode dispersion measurement based on continuous polarization modulation

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A novel technique is proposed for measurement of the group delay (GD) and the differential group delay (DGD) of optical material, components, and fiber. This new method is based on continuous polarization modulation of the stimulus optical field as opposed to sequential polarization state switching used in the traditional phase shift method. A new complete derivation of the phase shift method, based on the modified Jones calculus of elementary matrices, is presented. The derivation reveals that the phase shift measurement actually depends on all eight elementary parameters that represent DGD and optical frequency derivatives of polarization-dependent loss (PDL). Thus, the new expression for the phase shift includes the combined effect of PDL and DGD. The proposed method is evaluated by measuring a section of polarization-maintaining fiber and a 50 km length of single-mode fiber over a wavelength range from 1530 to 1610 nm. Measurements of DGD in a long single-mode fiber are shown to be highly insensitive to environmentally induced GD drift. © 2008 Optical Society of America

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1. Introduction

One of the key factors limiting the rate of data transmission in fiber optic communication networks is dispersion. The two types of dispersion in transmission networks are chromatic dispersion and polarization mode dispersion (PMD). Chromatic dispersion describes the dependence of the light's group velocity on wavelength; PMD describes the dependence on the polarization state. While chromatic dispersion can be easily compensated by using dispersion compensators, the compensation of PMD is more challenging because of its time-varying nature. Often, PMD sets a limit for rate of data transmission in optical fiber. In a single-mode (SM) fiber, the dependence of light propagation on the polarization state is due to birefringence induced by fiber geometry, local stresses, and fiber bending. Due to the random nature of the induced birefringence, the combined effect

is time and wavelength dependent. However, for each wavelength, the instantaneous birefringence is well defined and characterized by a fast and a slow axis that define two propagation modes. The group-delay (GD) difference between the slow and the fast propagation modes is known as a differential group delay (DGD). Since PMD in SM fiber is a manifestation of the distributed random birefringence, it evolves over time and environment because of the vibration, pressure, and thermally induced stresses. Maintaining low PMD and monitoring its evolution is essential in operating high-data-rate optical communication networks. Dispersion measurements are critical in qualifying optical fiber and components for use in those networks.

A very simple and precise method of measuring GD is the so-called phase shift method [1]. In the phase shift method, an intensity-modulated optical wave is launched into a device under test (DUT). Typically, the DUT is an optical fiber or an optical component. The intensity is detected at the device output, and the modulation phase is measured. The phase shift

of the detected intensity modulation is proportional to the GD of the DUT. Thus, if a tunable laser is used in the experiment, the GD can be measured over wavelengths allowing for determination of chromatic dispersion. The change in the GD can also be observed for different polarization states. By measuring the GD for four different polarization states, the DGD is determined [2,3]. The conceptual illustration of this technique is shown in Fig. 1(a). The procedure of this technique is similar to the measurement of insertion loss and polarization-dependent loss (PDL) [4]. Therefore, measurements of PMD and PDL are typically performed together. There are, however, some error sources that may affect the accuracy and, in some cases, the validity of the DGD measurements. The most obvious error source is related to the sequential switching of the input polarization states. The GD in a long length of fiber is never constant; it always drifts with environmental changes such as varying temperature or pressure. Thus, the sequential measurements of GD, as required to determine DGD, are affected by this drift. This effect is very difficult to suppress, since in long fiber lengths the GD is very large (e.g., $240\ \mu\text{s}$ in 50 km of SM fiber) while the DGD can be very small (e.g., 0.5 ps in 50 km of SM fiber). Sometimes additional measurements are taken to compensate for an assumed linear drift; however, the assumption of linear drift is often incorrect. An additional error occurs when the wavelength of the laser source is swept. In this case, any inconsistencies between wavelength sweeps convert GD variations that are due to chromatic dispersion into an apparent PMD. In a similar way, there is also a conversion of sharp spectral GD features into errors in measured DGD. Yet another error source that is typically omitted arises when the optical stimulus has frequency chirp in addition to the usual intensity modulation. The frequency chirp creates an error in measured DGD caused by the coupling of chirp with PDL as is shown in Section 2. This error source was not accounted for in the original analysis of the phase shift method [2,3].

In the method presented in this paper, a continuous polarization modulation of the stimulus optical field is used instead of the usual sequential polarization states switching. If the DUT has PMD, the modulation of the polarization state results in the modulation of the phase of the intensity modulation. It will be shown in the following sections that simultaneous observation of the harmonics of the demodulated phase provides all the measurement information required to solve for the DUT DGD. Thus, the solution is found in the frequency domain during a single measurement. The conceptual illustration of this new technique is shown in Fig. 1(b). In the proposed method, DGD measurement errors related to fiber GD drift or laser wavelength sweep inconsistencies are largely suppressed, since only a single wavelength scan of the stimulus laser is required.

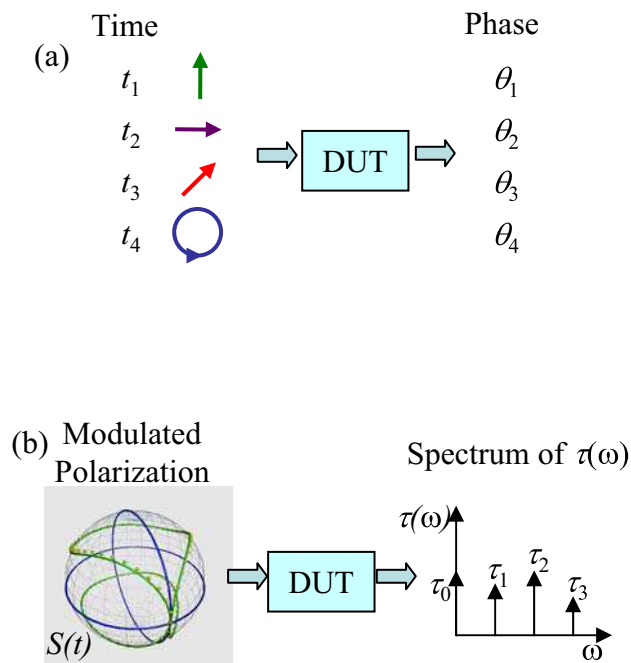


Fig. 1. (Color online) Conceptual illustration of the phase shift method. (a) Sequential polarization state switching. (b) Continuous polarization modulation with simultaneous observation of multiple harmonics of phase (delay).

In this paper we also present a new derivation of the phase shift method that is based on elementary matrices [5,6]. The elementary matrices, as described in [6], are ideal in the analysis of the measurement systems that contain modulated signals. They provide a simple means of accounting for amplitude and phase characteristics of optical components. The characteristics accounted for are the optical frequency derivatives of the amplitude and phase of the transmitted or reflected DUT signals, i.e., the derivatives of insertion loss and PDL, GD, and DGD. This constitutes a new concise and clear mathematical description of the phase shift method in terms of the elementary parameters and demonstrates, in a very elegant way, the combined effect of PDL and PMD [7].

2. Elementary Matrix Description of the Experimental Setup

The measurement system is illustrated in Fig. 2. The tunable laser source provides a monochromatic optical signal at the optical angular frequency ω_0 to the

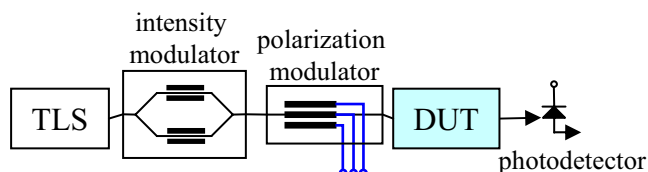


Fig. 2. (Color online) Simplified diagram of the experimental setup. TLS, tunable laser source.

integrated LiNbO₃ planar Mach–Zehnder intensity modulator. We initially assume that the intensity modulator is driven in a way that allows the existence of chirp; i.e., the modulation signals of the upper and lower arms of the Mach–Zehnder modulator have an in-phase component. All the sidebands created by the intensity modulation are assumed to be in the same polarization state until they propagate through the DUT. Therefore, we use a scalar notation for all the optical waves in the system until the DUT is reached. The optical wave in the upper arm of the Mach–Zehnder modulator is

$$e_U = \exp[j\omega_o t + j\psi + j(a - b) \cos(\omega_e t)]. \quad (1)$$

The optical wave in the lower arm is

$$e_L = \exp[j\omega_o t - j\psi - j(a + b) \cos(\omega_e t)]. \quad (2)$$

The phase term ψ accounts for the phase difference between the arms, a is the modulation index that is opposite between the arms, b is the modulation index that is common to both arms, and ω_e is the angular electrical frequency of the intensity modulation. The combined optical field is

$$e_U + e_L = 2 \exp[j\omega_o t - jb \cos(\omega_e t)] \cos[\psi + a \cos(\omega_e t)]. \quad (3)$$

Thus, ψ defines a bias point of the Mach–Zehnder modulator, a is the amplitude modulation index, and b is the phase modulation index. By including both indices we are able to identify their roles in our measurement system. Based on the Jacobi–Anger expansion [8], Eqs. (1) and (2) can be rewritten in terms of the individual sidebands. In particular, the modulation sideband n of Eq. (1) can be expressed as

$$e_U(n) = j^n J_n(a - b) \exp(jn\omega_e + j\psi). \quad (4)$$

Similarly, the modulation sideband n of Eq. (2) can be expressed as

$$e_L(n) = (-j)^n J_n(a + b) \exp(jn\omega_e - j\psi). \quad (5)$$

In the preceding equations, the sideband number n can take positive and negative values. The optical frequency term $\exp(j\omega_o t)$ was omitted for simplicity.

The polarization state of all the sidebands produced within the intensity modulator is altered in the same way by the polarization modulator. However, the DUT introduces slight differences between the sidebands that are dependent on the elementary matrices listed in Table 1 and described in detail in [6]. In considering propagation through the DUT we follow the procedure that was outlined in [6]. Specifically, we require that the perturbations be opposite in sign for positive and negative sidebands and increase with frequency deviation of the sideband from

Table 1. Elementary Matrices

Type	Phase Matrices	Amplitude Matrices
0° Linear	$\Theta_0 = p_0 \begin{pmatrix} -j & 0 \\ 0 & -j \end{pmatrix}$	$\Theta_4 = p_4 \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
	$\Theta_1 = p_1 \begin{pmatrix} j & 0 \\ 0 & -j \end{pmatrix}$	$\Theta_5 = p_5 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
45° Linear	$\Theta_2 = p_2 \begin{pmatrix} 0 & j \\ j & 0 \end{pmatrix}$	$\Theta_6 = p_6 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
	$\Theta_3 = p_3 \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$\Theta_7 = p_7 \begin{pmatrix} 0 & -j \\ j & 0 \end{pmatrix}$

the carrier frequency. The process is captured by the following equation:

$$\mathbf{E} = \sum_{n=-N}^N (\mathbf{I} + n\omega_e \mathbf{N}) \mathbf{A} [e_U(n) + e_L(n)], \quad (6)$$

where $\mathbf{E}$ represents the Jones vector at the output of the DUT, $\mathbf{I}$ is the identity matrix, $\mathbf{N} = \sum_i \Theta_i$ is an N matrix [5] that is represented by a sum of all eight elementary matrices shown in Table 1, and $\mathbf{A}$ is the Jones vector that represents an arbitrary polarization state taken by the optical carrier wave after propagation through the DUT. As mentioned above, the elementary matrices give rise to the small perturbations from the polarization state $\mathbf{A}$ that differ for different sidebands. It is important to realize that we were able to define the Jones vector $\mathbf{E}$ without careful accounting for all polarization transformations within the system. The arbitrary polarization state $\mathbf{A}$ can be described by the following equation:

$$\mathbf{A} = \begin{pmatrix} \cos(\alpha) \\ \exp(j\varphi) \sin(\alpha) \end{pmatrix}, \quad (7)$$

where the angles α and φ are parameters that uniquely define that polarization state. The corresponding Stokes vector is

$$\mathbf{S} = \begin{pmatrix} 1 \\ q \\ u \\ v \end{pmatrix} = \begin{pmatrix} 1 \\ \cos(2\alpha) \\ \sin(2\alpha) \cos(\varphi) \\ \sin(2\alpha) \sin(\varphi) \end{pmatrix}, \quad (8)$$

where q , u , and v represent 0° linear, 45° linear, and circular components of the vector.

The intensity at the detector is found from the Jones vector $\mathbf{E}$ that represents the sum of all field components after the DUT:

$$i = \mathbf{E}^* \mathbf{E}^T. \quad (9)$$

In the preceding equations T denotes the transpose and $*$ denotes the complex conjugate. The detected current (signal) is proportional to Eq. (9) by way of the photodetector responsivity.

The signal (9) is synchronously demodulated to determine its phase. The synchronous demodulation is

performed directly on the detected signal or after the detected signal is mixed down to a frequency that is suitable for digitizing and further digital processing. In our setup, the latter approach is used; however, mathematically, both approaches are equivalent. Synchronous demodulation results in an in-phase signal x and a quadrature signal y . The phase θ is found from the equation $\theta = \arctan(x, y)$. For small quadrature signal y , the equation simplifies to $\theta = y/x$. The GD τ is found by dividing the phase θ by the angular modulation frequency ω_c . The above procedure results in the following equation:

$$\tau = p_0 + p_1q + p_2u + p_3v + \frac{b}{J_1(2a)}(p_4 + p_5q + p_6u + p_7v). \quad (10)$$

The above equation is our main theoretical result. Some additional details are provided in the Appendix. The obtained result is substantially simpler and more general than the results previously reported [2,3]. Equation (10) elegantly captures the relationship between the components of the Stokes vector, q , u , and v , and all eight elementary parameters. Equation (10) clearly indicates that in the case of chirp-free intensity modulation ($b = 0$) the measured GD depends linearly on the elementary parameters p_0 , p_1 , p_2 , and p_3 only. The parameter p_0 represents the GD, while parameters p_1 , p_2 , and p_3 represent components of the DGD. Therefore, by considering four known input polarization states and by measuring four respective GDs τ_i , the set of four linear equations can be formed and solved, as illustrated in the next section, for GD (p_0) and DGD components (p_1 , p_2 , and p_3). This property was properly deduced [2,3]; however, the direct linkage with elementary matrices introduced by Jones [5] was not realized. Because of the simplifying assumption of the previous derivations [2,3], the complications resulting from the chirped laser source went unnoticed. As clearly shown by Eq. (10), in the presence of chirp ($b \neq 0$) all eight elementary parameters affect the measurement of the GD. Then, the optical frequency derivatives of the insertion loss (p_4) and the PDL (p_5 , p_6 , and p_7) may also look like GD or DGD, respectively. Thus, the measurement of pure GD requires chirp-free intensity modulation. However, if the laser sources used in the real optical network are not chirp free, the effects of loss and PDL on the perceived GD and DGD also have to be considered. Therefore, characterization of all eight elementary parameters is of significant importance in the metrology of optical networks and the optical components comprising those networks. Equation (10) provides the theoretical foundation for this complete characterization. It is important to realize that findings of this paper are in agreement with previous reports concerning the detrimental aspects of the combined effect of PMD and PDL [7].

3. Polarization Modulation

In this section we assume that the intensity modulation is chirpless. Then, the GD Eq. (10) becomes

$$\tau = p_0 + p_1q + p_2u + p_3v. \quad (11)$$

Note that the preceding equation has a form identical to the Mueller-matrix-based equation for transmitted optical power [4]:

$$p = m_{11} + m_{12}q + m_{13}u + m_{13}v, \quad (12)$$

where p is the fraction of the transmitted power and m_{1i} are the elements of the first row of the Mueller matrix. Both of these equations are typically solved by considering four different input polarization states. Instead of applying four polarization states, we propose a method based on continuous polarization modulation [9]. In this section, we provide the theoretical basis for this approach.

Consider the Stokes vector Eq. (8), where both polarization parameters α and φ are linearly changing with time at some rate ω , i.e., $\alpha = \varphi = \omega t$. Then, Eq. (8) becomes

$$\mathbf{S} = \begin{pmatrix} 1 \\ \cos(2\omega t) \\ 1/2 \sin(\omega t) + 1/2 \sin(3\omega t) \\ -1/2 \cos(\omega t) + 1/2 \cos(3\omega t) \end{pmatrix}. \quad (13)$$

The 0° linear component, q , of the Stokes vector contains the second harmonic of the modulation frequency ω , while the 45° linear component and the circular component (u and v) contain the first and the third harmonics of the modulation frequency. Consequently, if the elementary parameters p_1 , p_2 , and p_3 are nonzero, the same harmonics are detectable in the measured delay τ of Eq. (11). This behavior simply implies that, due to a nonzero DGD, the instantaneous GD depends on the input polarization state that evolves in time. The same behavior can be observed for arbitrary harmonics of the Stokes vector components. Since periodically varying components of the Stokes vector can be represented by a Fourier series, we can use the notation

$$\begin{pmatrix} 1 \\ q(t) \\ u(t) \\ v(t) \end{pmatrix} = \begin{pmatrix} 1 \\ q_0 + q_1 + q_2 + q_3 + \cdots \\ u_0 + u_1 + u_2 + u_3 + \cdots \\ v_0 + v_1 + v_2 + v_3 + \cdots \end{pmatrix}, \quad (14)$$

where subscripts denote the individual harmonics. Since light in the proposed method is highly monochromatic, the Stokes vector components satisfy the condition $q(t)^2 + u(t)^2 + v(t)^2 = 1$. To maintain the simplicity of notation, each of the indexed components of Eq. (14) is assumed to be complex, i.e., it contains both amplitude and phase information. For the nonzero elementary parameters p_1 , p_2 , and p_3 , each harmonic of the measured delay $\tau(t)$ is a linear

combination of corresponding harmonics contained in the modulated components of the Stokes vector. This can be captured by a simple matrix equation. For example, for the first three harmonics,

$$\begin{pmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{pmatrix} = \begin{pmatrix} q_1 & u_1 & v_1 \\ q_2 & u_2 & v_2 \\ q_3 & u_3 & v_3 \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix}, \quad (15)$$

where τ_i represent harmonics of $\tau(t)$, and the 3×3 matrix comprises harmonics of $q(t)$, $u(t)$, and $v(t)$. The 3×3 matrix describes polarization modulation. Therefore, if the polarization modulation is known, and the harmonics τ_i are measured, the elementary parameters that represent the components of DGD can be found. The DGD is found as $\tau_{\text{DGD}} = \sqrt{p_1^2 + p_2^2 + p_3^2}$. Unlike the traditional method requiring sequential polarization state switching [2], the variables required to solve for DGD are measured simultaneously in the frequency domain. Therefore, the effect of drift is to a large degree suppressed; all variables can be found within a single cycle of the polarization modulation. For lithium niobate x -cut polarization modulators, the modulation frequency can readily exceed 1 MHz (1 μ s polarization evolution).

The remaining parameter that needs to be found is the GD p_0 . The simplest way to solve for p_0 is to select polarization modulation such that $q_0 = u_0 = v_0 = 0$. This corresponds to modulation that produces pseudodepolarized light, i.e., light that on average does not have a preferred polarization state. Then, based on Eqs. (11) and (14), the DC component, τ_0 , of the measured delay $\tau(t)$ is equal to the GD:

$$\tau_0 = p_0. \quad (16)$$

The pseudodepolarized light makes the measurement of the DGD independent of the orientation of the DGD vector. Furthermore, unlike the traditional method [2], the measurements of the GD and DGD are decoupled. Therefore, measurements of small DGD values are not influenced by large GD values.

There are many schemes for modulating the polarization state [10]. In general, one should strive to maximize the determinant of the modulation matrix of Eq. (15). This maximizes the sensitivity of the measurement. For a well-defined matrix, the curve traced on the Poincare sphere captures a large volume. In the experiments reported here, we used the trajectory shown in Fig. 3, which has a characteristic figure 8 topology on the Poincare sphere. A similar topology is provided by the trajectory described by Eq. (13). The trajectory was traced at the modulation frequency of 1 kHz. The modulation matrix of Eq. (15) was measured experimentally. The detailed description of x -cut, z -propagating lithium niobate polarization modulators is beyond the scope of this paper. However, the details are found in papers on physics

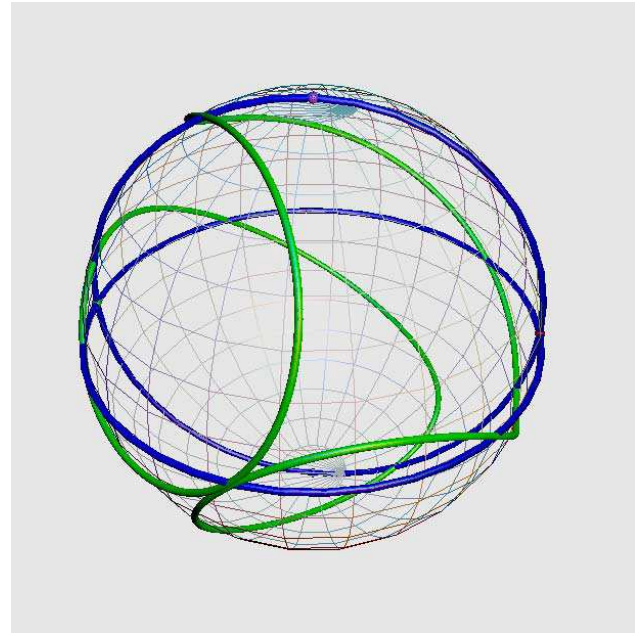


Fig. 3. (Color online) Poincare sphere representation of the polarization modulation trajectory.

of the modulators [11] and their characterization [12]. The basic polarization modulator, as shown in Fig. 2, has three electrodes. Typically the center electrode is held at ground potential, while two voltages are applied to the outer electrodes. By using two external voltages both the value and the angular orientation of the linear birefringence that is induced within the modulator can be controlled. For a fixed input polarization state, the induced birefringence uniquely determines the transformation of the polarization state. Therefore, by determining the relationship between the voltages and the induced linear birefringence one can produce a trajectory as described by Eq. (13) or any other trajectory.

4. Experimental Results

The proposed method was evaluated with two types of device exhibiting DGD. The first device was a length of a polarization-maintaining (PM) fiber; the second device comprised two spools of SM fiber (SMF-28) with a total length of 50 km. The respective DGD test results are shown in Figs. 4 and 5. Both tests were performed for a tunable laser sweep rate of 0.5 nm/s; the averaging window was about 100 ms, which corresponds to 100 cycles of polarization evolution at a polarization modulation frequency of 1 kHz. The averaging window of 100 ms at the wavelength sweep rate of 0.5 nm/s corresponds to a wavelength resolution of 50 pm. In the experiment we used an Agilent 81640B tunable laser, an Avanex low-chirp intensity modulator, and an EOSpace x -cut, z -propagating lithium niobate polarization controller. The intensity modulation frequency was 1 GHz. The speed of the measurement can be substantially improved by using higher-frequency polarization modulation.

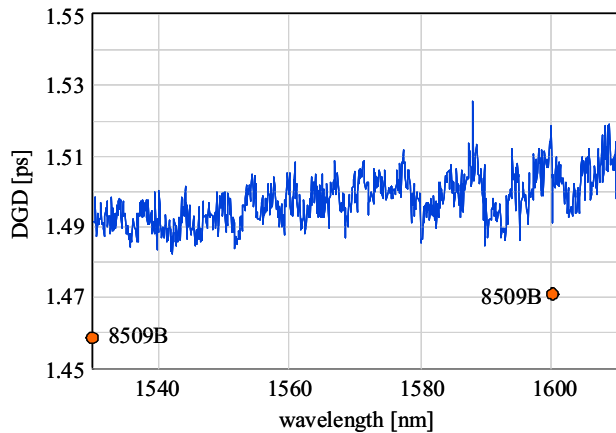


Fig. 4. (Color online) DGD measurements of a PM fiber patch cord.

The test of the PM fiber section verified that the test setup from Fig. 2 was properly calibrated. The PM fiber was first tested by using the classical polarimetric method at wavelengths of 1530 and 1600 nm. The DGD was estimated from the rotation of the polarization state on the Poincare sphere for a stepped wavelength of the tunable laser. The two discrete polarimetric measurements are shown in Fig. 4 as filled circles. Next, the same fiber was measured by using the new polarization modulation method over the wavelength range from 1530 to 1600 nm. As shown in Fig. 4 the new method differs from the polarimetric method by only about 2%. The difference could be caused by imperfect calibration of the modulation matrix or by existence of some polarized light in the system. The DGD noise is about 10 fs with 10–20 fs variations of DGD over wavelength. Some of the variations are real; some are due to the calibration that was performed over 10 nm windows. Overall, the results are very satisfactory, showing that the system is readily calibrated over wavelength and has good accuracy and that the noise levels are very low.

Next, the more challenging measurement of 50 km of SM fiber is shown in Fig. 5. The test is much more challenging because of the long fiber drifts in response to the environmental changes in temperature or pressure. Also, the GD has strong dependence on wavelength due to chromatic dispersion. This can introduce large errors in swept systems with polarization switching. In our experiment, the GD was observed to drift at about 8 ps/min while the GD variations over wavelength due to dispersion were about 1 ps/pm. In the systems that use multiple sweeps and polarization switching, 8 ps/min drift corresponds to DGD measurement errors of 133 fs for a 1 s sweep time. These errors can be reduced by assuming a linear drift and subtracting it; however, this technique does not always work. The DGD error due to chromatic dispersion can exceed 1 ps if wavelength repeatability between the sweeps is poorer than 1 pm. In multisweep systems the described error sources cause the increased DGD noise

and lack of repeatability. From our experience, for 50 km of SM fiber, the measurement noise in a multi-sweep method may exceed 100 fs, while trace-to-trace repeatability is poorer than 100 fs. Nevertheless, in the proposed new method, as shown in Figs. 4 and 5, we do not observe the increase of the DGD noise when the length of the DUT is increased from 1 m (PM patch cord) to 50 km (SM fiber). The noise level still remains at the level of 10 fs. Furthermore, the test repeated every few minutes, as shown in Fig. 5(a), reproduces the same DGD wavelength profile, showing excellent trace-to-trace repeatability. This demonstrates the stability of the new measurement method and its potential accuracy of 10 fs. Therefore, the new single-sweep implementation of the phase shift method offers substantial improvement in performance over a multisweep approach. The new technique, in its nature, is similar to the techniques that rely on direct observations of polarization state evolution [13–15]. This provides yet another explanation of the inherent immunity to environmentally induced drifts.

Since the DGD noise is reduced to about 10 fs and consecutive measurements are very repeatable, it is possible to observe the temporal behavior of SM fiber birefringence as illustrated in Fig. 5. Figure 5(a) shows that the behavior of the random birefringence

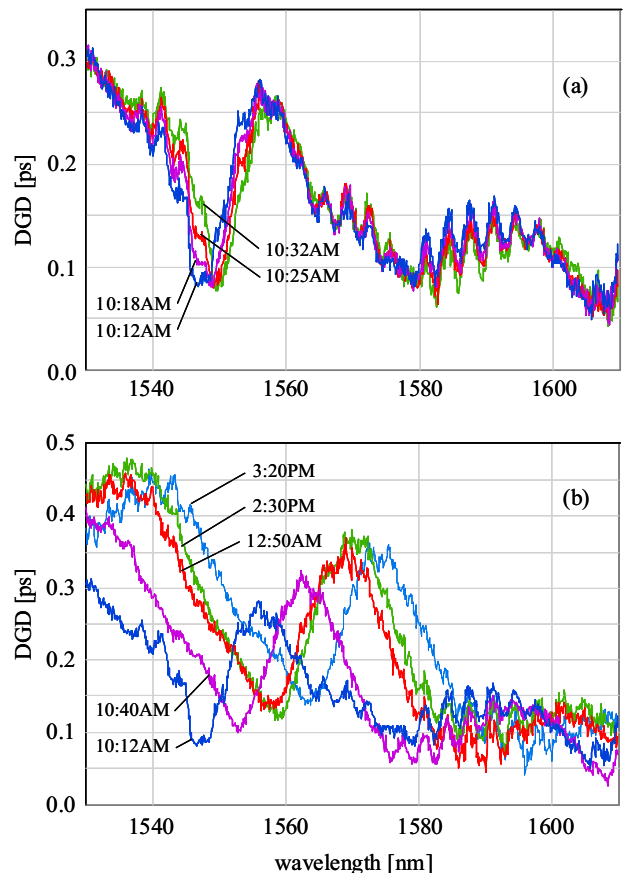


Fig. 5. (Color online) DGD measurements of a 50 km length of SM optical fiber: (a) short-term drift and (b) long-term drift.

induced within the SM fiber is quite stable in the laboratory environment. The only feature that shows some drift is in the neighborhood of 1550 nm. The test over several hours, shown in Fig. 5(b), shows some drift of DGD; however, the same features are definitely visible throughout the duration of the long-term test. Therefore, the test reveals behavior of the random birefringence within the SM fiber that, to our knowledge, was never observed with this kind of precision by using a multisweep technique according to [2].

5. Conclusions

The traditional phase shift method was derived based on the modified Jones calculus of elementary matrices. It was proved that the GD and the DGD, as measured in this technique, can be affected by insertion loss and PDL optical frequency derivatives. The effect takes place when the intensity modulation is not chirp free. This new result introduces

are used, the derivation is rather straightforward. Even in this simplified case, the form of Eq. (10) is properly captured, and the dependence on all eight elementary parameters is properly uncovered. For a single pair of sidebands, the multiplier of $(p_4 + p_5q + p_6u + p_7v)$ from Eq. (10) is

$$\frac{J_0(a+b)J_1(a+b) - J_0(a-b)J_1(a-b)}{J_0(a+b)J_1(a-b) + J_1(a+b)J_0(a-b)}. \quad (\text{A1})$$

For chirpless modulation ($b = 0$) the above multiplier is equal to zero. This implies that the measurement of phase is independent of the amplitude derivatives p_4 , p_5 , p_6 , and p_7 . As more sidebands are considered, the expression for the multiplier of $(p_4 + p_5q + p_6u + p_7v)$ becomes more complicated; however, there is a clear, easily identifiable pattern of Bessel function products in the numerator and denominator of the expression. For n pairs of sidebands, the expression for the multiplier is

$$\frac{\sum_{s=0}^{n-1} (2s+1) [J_s(a+b)J_{s+1}(a+b) - J_s(a-b)J_{s+1}(a-b)]}{\sum_{s=-(n-1)}^n J_s(a-b)J_{1-s}(a+b)}. \quad (\text{A2})$$

an important correction to the traditional phase shift method and accounts for the combined PDL and DGD. It also offers a theoretical basis for optical network characterization that accounts for the chirped optical sources that are used in the system.

The new formulation of the phase shift method was further used to introduce the continuous polarization modulation technique for measurement of GD and DGD. The proposed polarization modulation replaces the traditional polarization switching and suppresses error sources resulting from environmental drifts. The new measurement is performed on harmonics of the detected instantaneous phase (delay) that are simultaneously estimated in the frequency domain. The proposed method was experimentally verified on known birefringent PM fiber and evaluated on a 50 km length of SM fiber. For 50 km SM fiber, the new method showed DGD noise levels as low as 10 fs and similar repeatability. The new method allowed observation of the DGD behavior in the SM fiber spools in the laboratory environment, revealing slow drift of the DGD features. The proposed method can offer fast measurements of DGD by utilizing fast polarization modulation.

Appendix

The complexity of the derivation from Section 2 depends on the number of sidebands considered in Eq. (6). If only the carrier and two sidebands

To obtain the exact form of Eq. (10) all sidebands from Eq. (6) have to be considered. Then, the multiplier takes a simple form

$$\frac{b}{J_1(2a)}. \quad (\text{A3})$$

The numerator and the denominator of (A3) are found from the following Bessel function identities:

$$\sum_{s=0}^{\infty} (2s+1) [J_s(a+b)J_{s+1}(a+b) - J_s(a-b)J_{s+1}(a-b)] = b, \quad (\text{A4})$$

$$\begin{aligned} \sum_{s=-\infty}^{\infty} J_s(a-b)J_{1-s}(a+b) &= \sum_{s=-\infty}^{\infty} J_s(a+b)J_{1-s}(a-b) \\ &= J_1(2a). \end{aligned} \quad (\text{A5})$$

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